

STANDARD OPERATING PROCEDURE — EMERGENCY RESPONSE

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Operations Department | Approved: Operations Manager | Review Due: 2026-03-01 | Aligned with OSHA 29 CFR 1910.38

1. PURPOSE AND SCOPE

This SOP defines emergency response actions for unplanned equipment shutdowns, process upsets, and safety-critical events in Plant Areas A, B, and C. It provides first-response guidance to operators and technicians before specialist teams arrive. This document must be accessible at all operator workstations and field control panels. All operations staff must be trained and competency-assessed on this procedure annually.

2. GENERAL EMERGENCY RESPONSE PRINCIPLES

On detection of any emergency condition: STOP — THINK — ACT. Do not rush. A hasty response to an equipment emergency creates additional hazards. Assess the situation, ensure your own safety first, then initiate the appropriate response from this SOP.

- Personal safety is the first priority. Do not approach energised or hot equipment without PPE.
- Activate the local ESD system and notify the control room before any other action.
- Establish an exclusion zone of minimum 5 metres around any tripped rotating equipment.
- Do not attempt to restart tripped equipment without shift supervisor authorisation.
- Document all emergency events in the Shift Log within 30 minutes of occurrence.

3. EQUIPMENT-SPECIFIC EMERGENCY PROCEDURES

3.1 Pump-P3 — Emergency Trip Response

Pump-P3 is equipped with automatic trip protection on high bearing temperature (230 °F), high vibration (7.5 mm/s), and low suction pressure (< 5 psi). When Pump-P3 trips, the following response procedure applies:

1. Confirm trip annunciation at DCS and local panel (Panel C-08 red trip lamp).
2. Notify control room operator and shift supervisor immediately.
3. Identify and record the trip cause from DCS alarm history (Form OPS-EV-002).
4. Establish exclusion zone; place CAUTION barrier around Pump-P3.

For Pump-P3 post-trip access: a 5-minute cooldown period is sufficient for routine post-trip inspection checks by a qualified technician wearing appropriate PPE (heat-resistant gloves, face shield, and safety footwear). After 5 minutes, the technician may approach to check the bearing temperature gauge, inspect for visible seal leakage, and visually assess the pump for mechanical damage.

[NOTE] The 5-minute cooldown applies to EXTERNAL VISUAL INSPECTION ONLY. Any maintenance work requiring removal of bolted connections, drain plugs, or access covers requires a full lock-out/tag-out (LOTO) per SAFE-LOTO-007 and must be planned as a scheduled maintenance activity, not an emergency response.

5. Complete post-trip inspection report (Form EQ-PT-P3-001).
6. Consult Pump-P3 OEM manual for detailed root-cause guidance if bearing trip.
7. Restart only with Shift Supervisor written authorisation after clearing root cause.

3.2 Compressor-C1 — Emergency Trip Response

Compressor-C1 trips automatically on high discharge pressure (> 150 psi), high bearing temperature (> 250 °F), or high vibration (> 6.0 mm/s RMS). Following a trip, maintain a 10-metre exclusion zone. Do not approach until the unit has come to a complete stop (typically 3–5 minutes coast-down). Notify Engineering before any restart attempt.

3.3 Valve-V12 — Actuator Failure Response

If Valve-V12 fails to respond to open/close commands: switch to manual override on the actuator handwheel (clockwise = close). Do not force the handwheel if resistance exceeds 150 N·m — escalate to Mechanical Maintenance. Log the failure in the Valve Inspection Register.

4. ESCALATION AND NOTIFICATION MATRIX

All emergency events must be escalated per the following matrix. Time targets are from the moment the event is confirmed, not from when it is first detected.

- Within 5 min: Control room and Shift Supervisor notified.
- Within 15 min: Maintenance Supervisor notified if equipment repair required.
- Within 30 min: Plant Manager notified for any ESD activation.
- Within 1 hour: HSE Manager notified for any injury, fire, or release event.

- Within 24 hours: Incident Report Form INC-001 submitted to HSE Department.